



Dr. Zakaria SAWADOGO

Data Scientist — AI& Cybersecurity Researcher

*"Dare to invent the future."
Thomas Sankara*

Education

- 2020-2023 **PhD in Computer Science**, *AI & Cybersecurity Researcher* , Senegal, Gaston Berger University, Saint Louis
- 2014-2017 **Master in Software Engineering and Information systems**, *Joseph Ki-Zerbo University* , Burkina Faso, Ouagadougou
- 2010-2013 **Bachelor in computer Science**, *Joseph Ki-Zerbo University* , Burkina Faso, Ouagadougou
- 2009 **Baccalauréat - French equivalent to the Higher Leaving Certificate**, *Modern High School of Guiglo*, Côte d'Ivoire , Guiglo

Experience

- 2023 **Research Assistant in cyber security**, *Cyber Assurance Lab Internet of Things and Cyber Security Research Centre of Greenwich University,UK*, AI and ML, Based security, Malware detection and Threat modeling, I work with other researchers in the lab to develop techniques and tools to detect malware on Android using AI and formal methods in threat modeling
- 2020-2022 **Research Assistant in Data science and cyber security**, *Gaston Berger University, Senegal*, machine learning, deep-learning , I work on data collection, pre-processing, data mining, data visualization, and analysis for detecting Android malware.
- 11/2021- **Consultant Data Analyst**, *O VISION* , Cabinet de développement de solutions digitales, Consulting: Leveraging PowerBi for advanced data visualization and analysis

Burkina Faso – Ouagadougou

✉ zakaria.sawado@gmail.com • in [zakariasawado](#)

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- 2021-2024 **Sessional Lecturer**, *Higher Polytechnic School of Cheikh Anta Diop University of Dakar (UCAD)* , IT system security, Programming in python and Data visualization, Students develop academic projects in groups and I use Slack to communicate with the groups and Github or GitLab to track the progress and evaluate the contribution of each individual group member.
- 2021-2024 **Sessional Lecturer**, *Ecole Supérieure Multinationale des Télécommunications(ESMT) de Dakar, Sénégal* , Data visualization, Data mining and semantic technologies
- 2017–2020 **Computer Science Teacher**, *Joseph Ki-Zerbo University and Thomas Sankara University* , IT system security ,Algorithms and Python Programming, Students develop academic projects in groups and I use Slack to communicate with the groups and Github or GitLab to track the progress and evaluate the contribution of each individual group member.
- 2018-2019 **Junior Research Fellow**, *AI4D Africa*, Open Burkina , research program as part of an ambitious Artificial Intelligence for development project aimed at the preservation of indigenous / traditional languages

Professional development

- 2022 et 2024 **AfriConnect+** , *Training on connected objects* , Applied to, Agriculture and agricultural processing, Digital and artificial intelligence, Energy, housing and construction; , url:<https://esp.sn/evenement/africonnect-appel-a-projets/>
- 2020 **Certificate obtained** , *Data Science : Data analysis with Python* , Udemey Platform , url:<https://ichei.udemy.com/certificate/UC-c7be810d-d25d-4c19-9877-1156a4be96de/>
- 2020 **Certificate obtained** , *Artificial Intelligence for the Enterprise* , Udemey Platform ,
- 2019 **Certificate obtained** , *Deep Learning from A to Z* , Udemey Platform , url:<https://www.udemy.com/certificate/UC-91F8359P/>
- 2018 **Certificate obtained** , *Expert trainer in internet governance* , ISOC , <https://www.internetociety.org/fr/>
- 2016 **Certificate obtained** , *MOOC "Gestion de projet"* , école centrale de Lille , <http://certification.gestiondeprojet.pm/GdP8/a18761.pdf>

Conference and Vorkshop

- May 2025 **Co-author of the Digital Rights Manual for Beginners, a pedagogical guide published under the auspices of ENABEL.** , *The manual is designed to introduce young people, students, and citizens to the fundamental principles of digital rights, personal data protection, online safety, and digital inclusion* , It serves as a key reference resource for raising awareness about digital freedoms across Francophone Africa. The manual was officially launched at the 2025 edition of the Digital Rights and Inclusion Forum (DRIF) ,

- August 2024 **Expert on digital rights for ENABEL at the African Internet Freedom Forum 2024 in Dakar (September 23-27).** , *This experience enabled us to discuss critical issues surrounding internet governance and digital rights in Africa* , , As part of this, we worked on the development of a manual focusing on the digital rights of women and girls, aimed at making the internet safer, more inclusive and open for all
- October 2024 **Conducted a conference on Raising Awareness of Digital Rights for students in Koupéla and Koupéla as part of Digital Week 2024 on behalf of ENABEL.**, *I was in charge of the Introduction to Digital Rights and Personal Data Protection components.* ,
- 2023 **Organizer and project team leader of the Internet Society Burkina Chapter's Internet Security Forum** , *Burkina Faso chapter* ,
- 2020 2023 **I am a member of the Laboratoire d'Informatique, de Télécommunications et Applications (LITA) de l'ESP de Dakar**, , *which organizes editions 1 to 3 of the summer school on cybersecurity.* ,

Interests

- Professional Data science, Machine learning, deep learning, Internet Governance , software and network security
- Personal Football, cinema, debate and documentary

Awards

- 2022 **Best Paper Awards** , *The International Conference on Advanced Communications Technology (ICACT2022)* ,
- 2019 **Postgraduate Scholarship**, *PASET* , the Partnership for skills in Applied Sciences, Engineering and Technology , Regional Scholarship and Innovation Fund RSIF
- 2014 - 2017 **National Scholarship**, *University of Ouagadougou*, -, Burkina Faso Government
- 2013 **Excellence scholarship from Amici di Pietro Annigoni**, *University of Ouagadougou* , Burkina Faso,

Associative responsibility

- Vice president of Burkina Faso ISOC chapter, 2021-2024
- Vice president of IGF-Burkina, between February 2021-2023
- ISOC Fundamentals Coordinator Burkina Faso Chapter, 2019 and 2020
- President of the Computer Science Club of the University Joseph Ki-Zerbo, between April 2013 and April 2016

Publications

- [1] **Zakaria Sawadogo**, Babacar Thioye, Ibrahima Gaye, Jean-Marie Dembele, Gervais Mendy, and Samuel Ouya. Mid@ndro: a middleware architecture for malware detection on android. In *2023 IEEE 6th International Conference on Cloud Computing and Artificial Intelligence: Technologies and Applications (CloudTech)*, pages 1–5, 2023.
- [2] **Zakaria Sawadogo**, Muhammad Taimoor Khan, Jean Marie Dembelle, Gervais Mendy, and Samuel Ouya. Towards a deepmalob improvement in the use of formal security risk analysis methods. In *2023 IEEE 6th International Conference on Cloud Computing and Artificial Intelligence: Technologies and Applications (CloudTech)*, pages 1–5, 2023.
- [3] **Zakaria Sawadogo**, Jean-Marie Dembele, Gervais Mendy, and Samuel Ouya. Android malware detection: Feature update using incremental learning approach: Further investigation of ufila. In *2023 25th International Conference on Advanced Communication Technology (ICACT)*, pages 1446–1449, 2023.
- [4] **Zakaria Sawadogo**, Jean-Marie Dembele, Gervais Mendy, and Samuel Ouya. Android malware detection: An in-depth investigation of the impact of the use of imbalance datasets on the efficiency of machine learning models. In *2023 25th International Conference on Advanced Communication Technology (ICACT)*, pages 1460–1467, 2023.
- [5] **Zakaria Sawadogo**, Jean-marie Dembele, Gervais Mendy, and Samuel Ouya. Zero-Vuln : Using deep learning and zero-shot learning techniques to detect zero-day malwares Android. In *The 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, number July, 2023.
- [6] **Zakaria Sawadogo** , Jean Marie Dembele, Attoumane Tahar, Gervais Mendy, and Samuel Ouya. DeepMalOb: Deep Detection of Obfuscated Android Malware. *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST*, 459 LNICST:307–318, 2023.
- [7] **Zakaria Sawadogo** , Gervais Mendy, Jean Marie Dembelle, and Samuel Ouya. Android malware classification: Updating features through incremental learning approach(ufila). In *2022 24th International Conference on Advanced Communication Technology (ICACT)*, pages 544–550, 2022, **Best Paper**.
- [8] **Zakaria Sawadogo**, Gervais Mendy, Jean Marie Dembele, and Samuel Ouya. Android malware detection: Investigating the impact of imbalanced data-sets on the performance of machine learning models. In *2022 24th International Conference on Advanced Communication Technology (ICACT)*, pages 435–441, 2022.
- [9] Arthur D Sawadogo, **Zakaria Sawadogo** , Steve T M Ataky, Khalid M Askia, Kalmogo Roland, and Issa Boussim. Preliminary Investigation of Mobile Banking Attacks in West Africa: Feedback from Orange Money Customers in Burkina Faso. In Yahya H Sheikh, Idris A Rai, and Abubakar D Bakar, editors, *e-Infrastructure and e-Services for Developing Countries*, pages 15–24, Cham, 2022. Springer International Publishing.

- [10] Sankalp Ghatpande, Hadja Ouattara, Daouda Ahmat, **Zakaria Sawadogo** , and Tegawendé F. Bissyandé. Secure, transparent and uniform mobile money for internet-underserved areas using sporadically-synchronized blockchain. In Gervais Mendy, Samuel Ouya, Ibra Dioum, and Ousmane Thiaré, editors, *e-Infrastructure and e-Services for Developing Countries*, pages 120–130, Cham, 2019. Springer International Publishing.
- [11] Tegawendé F. Bissyandé, Jonathan Ouoba, Daouda Ahmat, A. D. Sawadogo, and **Zakaria Sawadogo** . Bootstrapping software engineering training in developing countries - return on experience at the university of ouagadougou. In *AFRICOMM*, 2014.

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